Holds, Then Snaps

How Cooperation Fails Under Execution Error in the Iterated Prisoner's Dilemma

Tarık Şahar

Istanbul Technical University, Data Science and Analytics

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Abstract

Cooperation under noise is widely described as eroding in proportion to how unreliable agents are. We find that it does not. Sweeping an execution error rate ε against a continuation probability w over a fifteen-strategy pool — 152 cells, five replicates each, 760 runs — and measuring cooperation from the moves actually played rather than from which strategies survive, normalised cooperation declines almost linearly to **0.788** at $\varepsilon = 0.28$ and then falls to **0.628**, **0.162** and 0.008 over the following three grid steps. Under the usual account we would expect a decline roughly tracking ε; we observe a plateau and then a cliff. At an error rate where more than a quarter of all moves are mistakes, cooperation still retains four fifths of what the noise leaves available; two steps later it retains none. The viable region is a staircase in which the two parameters trade against one another, and the trade is bounded: past $\varepsilon \approx 0.30$ no horizon buys cooperation back. Two predictions derived from the payoff structure before the sweep existed both hold — the basin of defection scales as $(1 - w)$ (log–log slope 1.046 against a derived 1.000), and the retaliator share required rises with ε (0.57% to 10.0% over $\varepsilon = 0$ to 0.08, monotonically). Because we found that removing a single strategy changes the tournament winner, we treat the roster as a measured variable rather than a setting: across 120 randomly drawn sub-rosters the dependence of the map on the cast falls from **0.177** at five strategies to **0.037** at thirteen, and the seven-strategy roster we used in the noiseless phases sits at the *minimum* of 24 draws at its own size. Three mechanisms for surviving a mistake are separable, and the strongest is dilution — judging an opponent by an accumulated record rather than by a single move — whose removal costs 0.10 of viable range against 0.06 for structural contrition and 0.00 for probabilistic forgiveness. Four claims advanced during this study were later overturned, in every case because a summary statistic had reported the state of the instrument rather than the state of the system. The fourth produced a conclusion that was wrong rather than imprecise; we report it as a documented negative result alongside the corrected estimate.

1 Introduction

A round-robin tournament between hand-coded strategies, followed by replicator dynamics on the resulting payoff matrix, reproduces results settled since Axelrod [1, 2]. Tit-for-Tat does well; unconditional defection eliminates itself once there is nothing left to exploit. Building that apparatus is a reasonable exercise, but it is not a finding.

We ask a narrower question the apparatus can answer: under what conditions does cooperation survive, and what does its failure look like? We vary two properties of the world. Agents sometimes play a move they did not intend, at rate ε. And the game is not of publicly known length: after each round it continues with probability w. The second matters for more than realism — with a fixed, commonly known horizon, backward induction makes defection the rational play in every round, which undermines the premise of the simulation.

We therefore treat the tournament and the population model as apparatus rather than result: they exist to earn the right to trust the code. Three questions follow. Where does the boundary of viable cooperation sit, what does the crossing look like, and how much of the answer belongs to the model rather than to the strategies that happened to be in the room?

2 Model

2.1 The game

Each round, two agents simultaneously choose to cooperate (C) or defect (D). Payoffs follow the canonical ordering $T > R > P > S$ with $2R > T + S$, the second condition ensuring alternating exploitation does not beat sustained mutual cooperation. Throughout, $T = 5$, $R = 3$, $P = 1$, $S = 0$ — Axelrod's values, so results are directly comparable with the literature, and treated as a default to vary rather than a constant.

Keeping $S = 0$ is deliberate rather than incidental. Adding a constant to all four payoffs leaves the game unchanged: the ordering, every best response and every matchup outcome are identical, and the tournament ranking is unaffected. What it changes is the speed of the replicator dynamics, because the discrete update is a ratio and inflating every payoff pushes that ratio towards one — raising all four payoffs by 100 leaves nothing extinct within 200 generations. Negative payoffs are a separate matter, taken up in Section 5.

2.2 Strategies

Strategies are pure functions of the match history: given both players' move sequences so far, return the next move. None of them learn; adaptation happens only through changes in population share.

The pool of fifteen is chosen by *mechanism* rather than by variety, so that a result can be attributed to a way of deciding rather than to a name: unconditional play (Always Cooperate, Always Defect); reciprocity at varying tolerance (Tit-for-Tat, Tit-for-Two-Tats, Two-Tits-for-Tat, Grim Trigger); error-aware play (Generous Tit-for-Tat [3], Contrite Tit-for-Tat [4], Pavlov [5]); an opening variant (Suspicious Tit-for-Tat); probing and exploitation (Prober, Gradual); aggregate history (Soft Majority); and two baselines (Random, Alternator).

Contrite Tit-for-Tat is the one entry designed for the question this report asks. It tracks whether it is in good standing, and when its own defection put it in bad standing it accepts the resulting punishment instead of answering it. Standing is defined on moves actually played, so an execution error — indistinguishable from a deliberate defection even to the player who made it — is still assigned correctly. In self-play at $\varepsilon = 0.02$ it scores 2.94 of a possible 3.00, where Tit-for-Tat scores 2.455.

2.3 Tournament

Every strategy plays every other and itself. The output that matters is not the leaderboard but the matrix M, where $M[i, j]$ is the mean payoff *per round* that i earns against j, with self-play on the diagonal. Per-round rather than per-match, so fitness does not scale with the horizon; with self-play included, because a population model needs the payoff a strategy earns when it meets a copy of itself. The leaderboard is the row mean of that same matrix, so the two cannot disagree.

Randomness is addressed rather than assumed. Each match draws from a generator keyed by its coordinates (i, j, trial) off one documented root seed, rather than from a running stream. Adding a strategy to the roster therefore cannot silently shift numbers elsewhere, and any single cell can be re-run in isolation. Within a match the two players draw from independent substreams, so a stochastic strategy in self-play does not mirror its own moves.

Under a stochastic horizon the per-round payoff is pooled as $\mathbb{E}[\text{total}]/\mathbb{E}[\text{length}]$ rather than averaged over per-match means, so a one-round match does not count as heavily as a hundred-round one. The two coincide at fixed length, which is why the distinction is easy to miss.

2.4 Replicator dynamics

Population shares evolve by $x_i(t+1) = x_i(t)\, f_i(t)/\bar{f}(t)$, where f_i is strategy i's expected payoff against the current mixture, taken directly from M [6, 7]. Dividing by mean fitness keeps the shares on the simplex by construction rather than by repair.

Two conventions are required and neither is given by the mathematics. Selection intensity enters as $f = (1 - s) + s \cdot \text{payoff}$, making the timescale an explicit choice rather than an accident of how the payoff numbers happen to be written. And shares approach zero without reaching it, so calling a strategy extinct requires a cutoff — applied only at reporting time and never inside the dynamics, for reasons given in Section 5.

3 Methods

3.1 Execution error and a stochastic horizon

ε is execution error: the intended move is flipped on the way out, and both histories record what was actually played, so the two players never disagree about what happened. This is error in doing, not error in seeing; the misperception case [8], in which the players hold different histories of the same match, is outside the scope of this report.

w gives a geometric horizon with expected length $1/(1 - w)$. The length cap is derived from w rather than fixed, because a fixed cap truncates the distribution unevenly: at $w = 0.99$ a cap of 200 rounds cuts 13% of the length distribution and pulls the mean from 100 to about 86. Deriving the cap from w removes the same negligible tail at every w.

Both parameters default to a no-op rather than to a neutral value: at $\varepsilon = 0$ and $w = 1$ the new code paths are skipped rather than executed with harmless arguments, so no draw is taken from a generator that was not taken before. All pre-existing tests passed unchanged and bit-identically before anything was built on top.

The grid is $\varepsilon \in [0, 0.35]$ against $w \in [0.5, 0.99]$: 19 error rates by 8 continuation probabilities, 152 cells, five replicates each and 760 runs in total, with the replicator run to convergence at $G = 60{,}000$.

3.2 Measuring cooperation from play, not from names

Cooperation is measured as the fraction of moves actually played as C. The obvious alternative — scoring a population cooperative when the strategies named as defectors are extinct — fails under noise, and fails in the direction that flatters the result (Section 5).

Because the observable rate is confined to $[\varepsilon, 1-\varepsilon]$ and that band narrows as ε grows, a fixed cutoff is a moving target against it. Results are therefore also reported normalised as $(\text{rate} - \varepsilon)/(1 - 2\varepsilon)$, where 1 is total cooperation and 0 total defection at every error rate. Both readings locate the same boundary, so the boundary is a property of the population and not of the cutoff.

3.3 The roster as a measured variable

$M[i, j]$ depends only on strategies i and j and on $(\text{payoffs}, \varepsilon, w, \text{rounds})$, never on who else is in the tournament. A sub-roster's matrix is therefore the corresponding submatrix of one already computed, and its map can be re-derived without re-simulating anything. Every roster comparison in this report is made against one shared set of sampled matrices, so a difference between two rosters cannot be sampling noise between two runs.

Two quantities follow. *Sensitivity* draws random sub-rosters at sizes 5, 7, 9, 11 and 13, 24 draws each, and reports the mean absolute difference between a drawn roster's map and the pool's. *Influence* removes each strategy in turn and reports both how far the map moves and how much viable range is lost.

3.4 Software

Python 3.14 with NumPy; Matplotlib for figures. The test suite is 200 tests, including five named invariants and a move-for-move cross-check against an independent implementation [9]. Data derivation and rendering are separate stages: `report_data.py` derives six CSVs from the raw grids and matrices, and `figures.py` renders each figure from those CSVs alone, with nothing hand-entered.

4 Results

4.1 The apparatus before noise

With no error and a fixed horizon the model reproduces what it should, and two results worth carrying forward fall out of the first run.

Grim Trigger tops the tournament at **2.713** against Tit-for-Tat's **2.612**. Subtracting the two rows opponent by opponent, however, puts the whole margin in a single column: the strategies are identical to 0.0000 against every opponent that is not a coin flip, and Grim beats Tit-for-Tat by **0.7090** against Random, which is the whole of the 0.1013 leaderboard gap once divided across seven opponents. The advantage is not better cooperation. Random defects early, tripping Grim's trigger and licensing it to harvest a coin-flipper for the remaining rounds. Remove Random and the ranking collapses into a three-way tie: Tit-for-Tat 2.6658, Grim Trigger 2.6658, Tit-for-Two-Tats 2.6650. The leaderboard reports its roster, not its winner.

Source: `results/phase_a_leaderboard.csv`

Under replicator dynamics from equal shares (Figure 1) the defectors eliminate themselves — Always Defect at generation 44, Random at 59 — because exploitation pays only while naive cooperators remain to exploit. What follows matters more than the eliminations. The five survivors all cooperate with one another, so their fitnesses are identical and the spread among them is exactly **0**: selection has nothing left to act on and stops. The final proportions record the race, not an equilibrium that was chosen. This is the sense in which no pure strategy is evolutionarily stable in the repeated game [10]: strategies that tie everywhere drift into one another for free. Neutral drift of this kind is not an artefact of a small roster — in heterogeneous populations it is the mechanism by which a reciprocator gives way to more generous strategies once it has done the work of clearing the defectors [11], and it is why the noiseless model cannot answer the question this report asks. Only an opponent that defects intermittently breaks the ties, which is exactly what $\varepsilon > 0$ supplies.

Sampling 1000 starting mixtures uniformly from the simplex separates the model from the starting point. The survivor set belongs to the model: Always Defect and Random die in **1000 runs out of 1000**. The proportions belong to the start — Grim Trigger finishes anywhere between 0.0005 and 0.9455 of the population depending only on where the run began. We therefore report no final proportion as a finding.

Two quantitative results also emerge, and both are predictions about a sweep we had not yet run. First, the retaliator share needed to defeat a defector majority scales as $1/N$, falling from 6.11% at $N = 10$ to 0.14% at $N = 400$: Always Defect's entire advantage over Tit-for-Tat is one round of exploitation worth $T - P$, amortised over N rounds. Since a match under continuation probability w lasts $1/(1-w)$ rounds in expectation, we predict cooperation's basin should shrink like $(1 - w)$. Second, what defeats defection is retaliation rather than cooperation: holding Always Defect at 99% and splitting the remaining 1% between Always Cooperate and Tit-for-Tat, **0.252**% Tit-for-Tat is enough to turn the population cooperative, while a population that is half Always Defect and half Always Cooperate needs *more*, **0.377**%. An Always Cooperate pays a defector T every round where a Tit-for-Tat pays P after the first; cooperators feed defectors and retaliators starve them. Since that argument rests on retaliators recognising one another, we predict the required share should rise with ε.

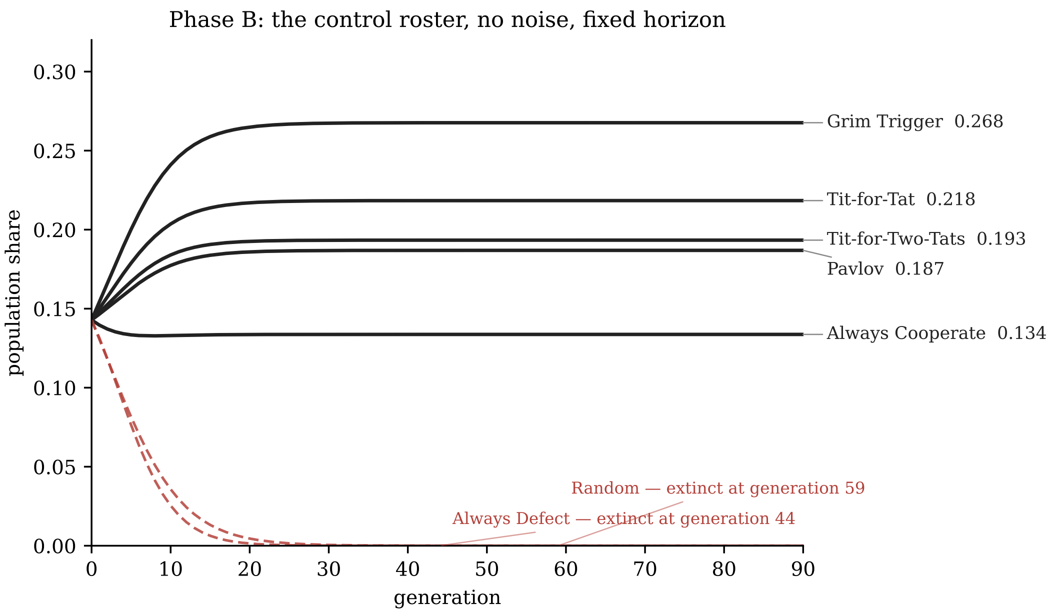

Figure 1: Replicator dynamics on the seven-strategy control roster with no execution error and a fixed horizon. Always Defect and Random are eliminated; the survivors are mutually neutral, so the final proportions record the transient rather than an equilibrium selection has chosen.

Source: `results/fig6_phase_b_trajectory.csv`

4.2 The map

Figure 2 gives the sweep. The viable region is a staircase, not a rectangle: the two parameters trade against one another, and more noise demands a longer shadow of the future. On the seven-strategy control the steps are sharp — $\varepsilon = 0$ survives from $w \geq 0.7$, ε up to 0.08 needs $w \geq 0.8$, $\varepsilon = 0.10$ and

0.12 need $w \geq 0.9$, $\varepsilon = 0.14$ needs $w \geq 0.98$, and above 0.14 nothing rescues it. The fifteen-strategy pool sustains cooperation to $\varepsilon = 0.30$.

The trade is bounded in both directions. No horizon buys cooperation back past the pool's ceiling, and no error rate below it is survivable at short horizons: at $w = 0.5$, where a match lasts two rounds in expectation, cooperation fails even at $\varepsilon = 0$.

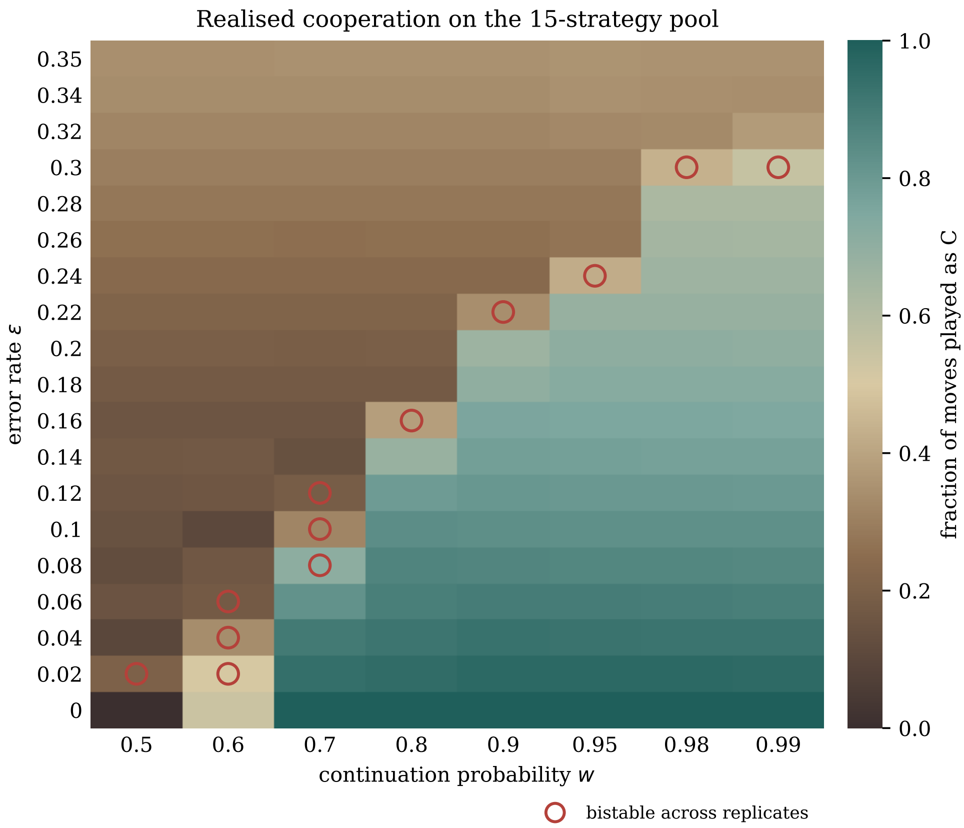

Figure 2: Realised cooperation across the error rate ε and the continuation probability w, on the fifteen-strategy pool. Colour is the fraction of moves actually played as C, averaged over five replicates. Ringed cells are bistable: replicates differing only in the tournament seed reach different attractors, and a mean describes neither.

Source: `results/fig1_map_pool.csv`

4.3 The edge

Figure 3 is the shape the report is named for. Along the longest horizon, normalised cooperation declines gently and almost linearly across two thirds of the range — 0.911 at $\varepsilon = 0.10$, 0.836 at 0.20, still **0.788** at $\varepsilon = 0.28$. At an error rate of 28%, where more than a quarter of all moves are mistakes, cooperation retains nearly four fifths of what the noise leaves available.

Then it goes: **0.628** at $\varepsilon = 0.30$, **0.162** at 0.32, 0.008 at 0.34. Three grid steps take it from most of the band to none of it. Under a model in which reliability and cooperation degrade together we would expect the middle of this curve to be its steepest part; we observe the opposite. The lower panel shows the same collapse from the other side — the realised rate descending onto the line rate $= \varepsilon$, the signature of a population in which no move is cooperative except by accident.

Naming a single ceiling is a convention, and one that discards the interesting part. The value at $\varepsilon = 0.30$ is the least trustworthy point on the curve: across replicates the realised rate there ranges from 0.312 to 0.613, the bistability of the boundary appearing as a wide spread rather than as a number. The three figures around the edge say what one figure cannot.

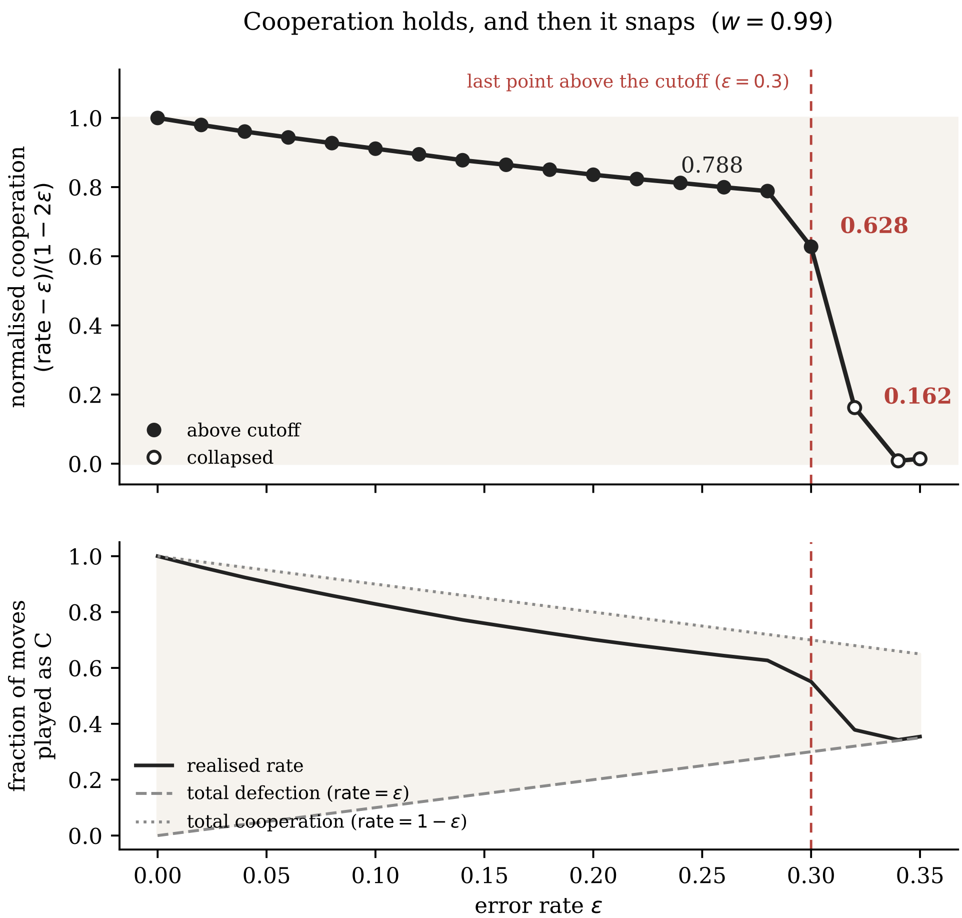

Figure 3: The shape the report is named for, along the longest horizon ($w = 0.99$). Upper panel: cooperation normalised onto the band the error rate leaves available, $(\text{rate} - \varepsilon)/(1 - 2\varepsilon)$, so that 1 is total cooperation and 0 total defection at every ε. It holds at 0.788 through $\varepsilon = 0.28$, then falls to 0.628 and 0.162 over the next two grid steps. Lower panel: the raw rate against the two bounds it lies between; collapse is the rate meeting the lower one.

Source: `results/fig2_edge_profile.csv`

4.4 Two predictions, tested

Both predictions in Section 4.1 were derived from the payoff structure before the sweep existed, which makes agreement worth more than the sweep alone.

Holding a population at 90% Always Defect and measuring the Tit-for-Tat share needed to save cooperation, the log–log slope against $(1-w)$ is 1.046 against a derived 1.000, and the ratio needed$/(1-w)$ varies by a factor of only 1.14 across $w = 0.9 \ldots 0.99$. The basin scales as $(1-w)$ as derived.

The required retaliator share rises with ε from 0.5675% to 10.0% over $\varepsilon = 0$ to 0.08 at $w = 0.99$, a factor of 17.6, monotonically — and more sharply than the prediction stated, because past $\varepsilon = 0.10$ the answer is not a larger number but no number: on that roster no retaliator share defeats a 90% defector majority.

4.5 The roster as a variable

Figure 4 reads the same sampled tournaments twice: over the seven strategies our noiseless phases used, and over the fifteen-strategy pool. The control breaks above $\varepsilon = \mathbf{0.14}$ where the pool sustains cooperation to **0.30**. This is not a better roster winning. It is a roster containing strategies designed for a world with mistakes, asked a question about mistakes.

Across 120 sub-rosters the dependence of the answer on the cast falls monotonically and steeply with roster size: the mean absolute difference between a drawn roster's map and the pool's is **0.177** at size five and **0.037** at size thirteen (Figure 5). The minimum ceiling rises with size — 0.06, 0.14, 0.16, 0.16, 0.20 — and our seven-strategy roster's 0.14 sits at the *minimum* of 24 draws at its own size, against a median of 0.26. We had assembled it for a world without mistakes, and on this question it was close to the worst cast available.

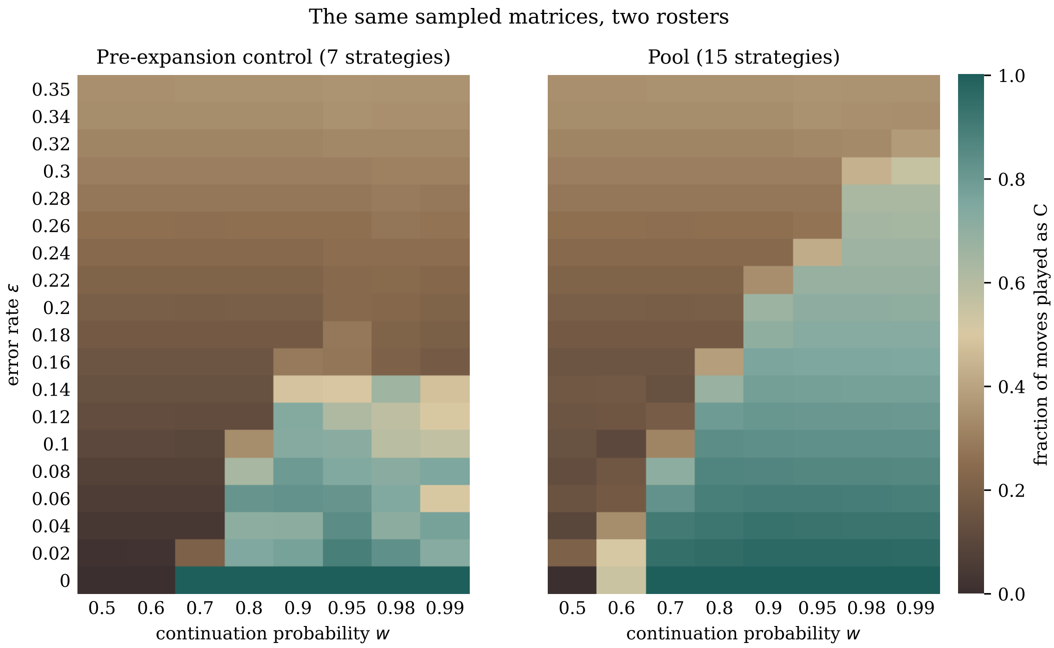

Figure 4: The same sampled tournaments read twice: once over the seven strategies the earlier phases used, once over the fifteen-strategy pool. Because both maps are submatrices of one set of matrices, the difference between them cannot be sampling noise between two runs. *Source:* `results/fig3_roster_comparison.csv`

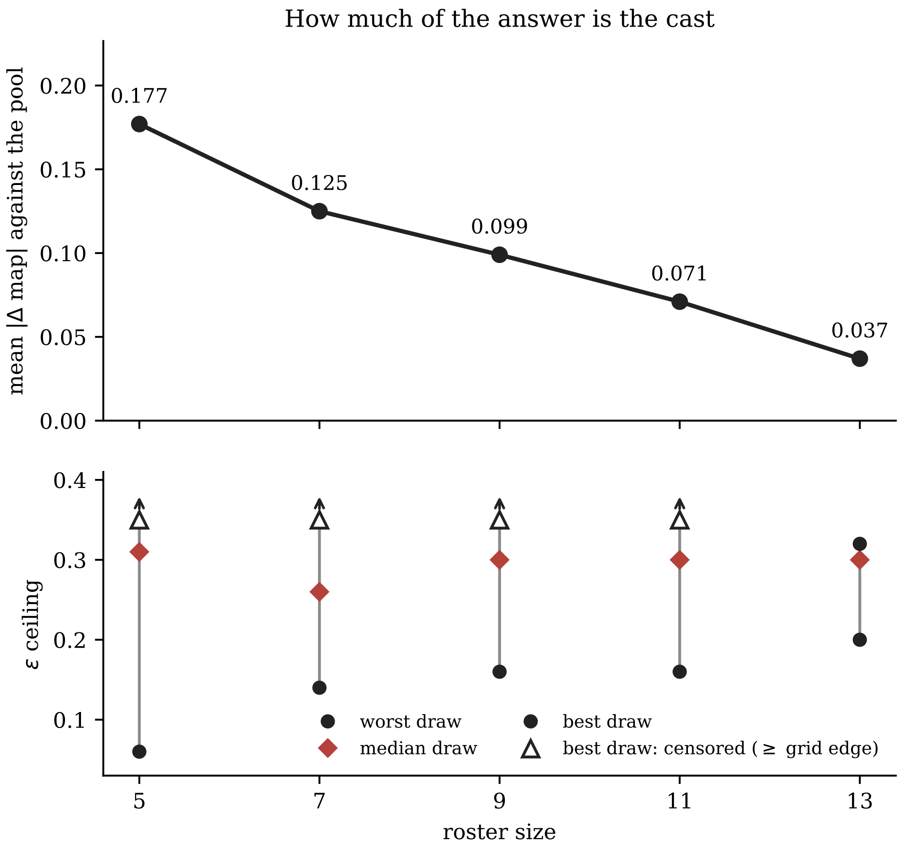

Figure 5: How much of the answer is the cast. Upper panel: the mean absolute difference between a drawn sub-roster's map and the pool's, against roster size. Lower panel: the spread of ε ceilings at each size. Open upward triangles mark censored values — rosters still cooperating in the last column of the grid, for which the measurement is a lower bound and not a location.

Source: `results/fig5_sensitivity.csv`

4.6 Influence, and three ways of surviving error

Removing each strategy in turn and re-deriving the whole map gives Figure 6, whose two panels disagree in a way that is the point of the figure. Always Defect moves the map more than anything else, by **0.101**, and costs **exactly zero** viable range: the antagonist shapes the landscape without setting its limit. Soft Majority is only third by map shift, at 0.054, and *first* by range lost — removing it drops the pool from 0.30 to **0.20**. Contrite Tit-for-Tat sits between them at 0.070 and 0.06. A ranking by either quantity alone would misdescribe the pool.

Leave-one-out does not compose, and treating it as though it did is the trap here. A strategy can measure as uninfluential precisely because something else covers its mechanism: Generous Tit-for-Tat ranks fourteenth of fifteen, not because forgiveness is irrelevant but because Contrite Tit-for-Tat and Soft Majority already handle error, and handle it better. Remove all three and error-handling leaves with them. We therefore measured trimmed rosters rather than reading one off the ranking. Keeping the twelve most influential moves the map by 0.0145 and changes the verdict in four cells of 152; keeping ten moves it by 0.0290 and changes nine; keeping eight moves it by 0.0381 and changes eight. All three hold the ε ceiling at 0.30, and all three sit below the 0.054 a change of random seed produces — which is exactly why the map-shift column cannot rank them. Nor does the w staircase separate ten from eight: against the pool, ten differs at nine of sixteen ε steps and eight at eight, in both cases by a single grid step. Twelve is the only trimmed roster that is cleanly better than the others on both counts, and the honest reading is that this grid can distinguish twelve from the pool but not ten from eight. We therefore report the pool and treat trimming as a robustness check, not as a recommended roster.

Mechanism	**Strategy**	**Map shift**	**Range lost**
Dilution	Soft Majority	0.054	0.10
Contrition	Contrite TFT	0.070	0.06
Forgiveness	Generous TFT	0.013	0.00

Table 1: Three mechanisms for surviving execution error, by how far the map moves and how much ε range is lost when each is removed from the pool. Computed at two replicates per cell, as every leave-one-out and sub-roster sweep is; see Section 5.

Source: `results/fig4_influence.csv`

The pool contains three distinct answers to the problem of a mistake (Table 1), and they are not equally good. **Contrition** answers a mistake: the strategy recognises that its own defection put it in the wrong and accepts the punishment rather than retaliating against it. **Forgiveness** ignores a fixed fraction of defections without asking whose fault they were — statistical rather than structural, and dominated by contrition, which asks. **Dilution** does not respond to the mistake at all: Soft Majority answers an opponent according to what they have done most often across the whole history, so a single slip barely moves the verdict.

Dilution wins. Judging someone by their record rather than by their last move is the most robust defence against noise the model contains. It is also the only one of the three requiring unbounded memory, which is a real cost and not a free lunch.

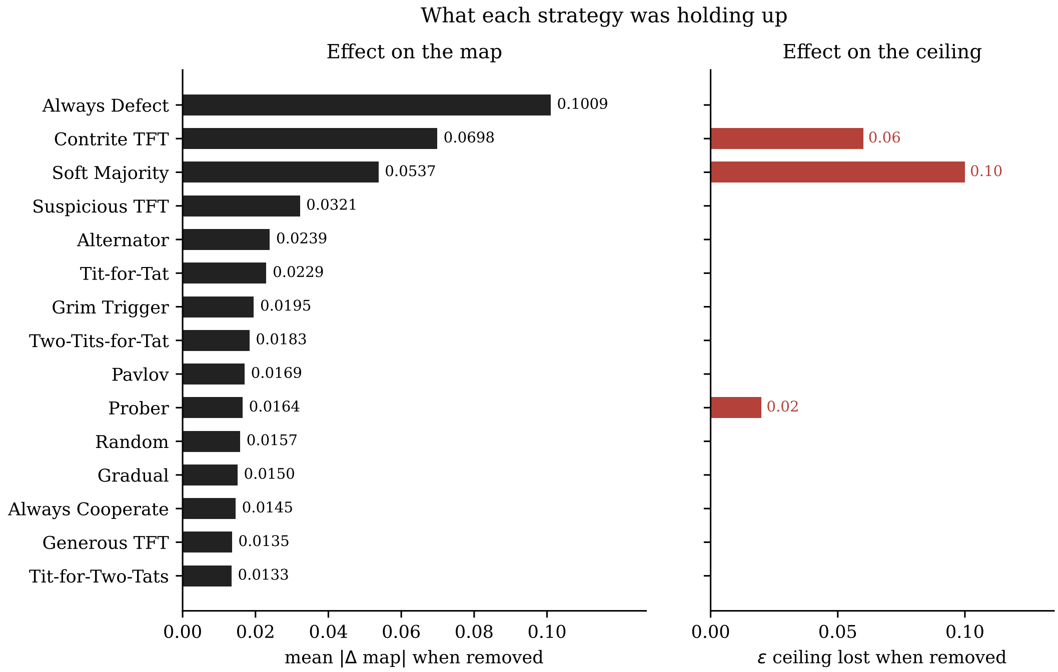

Figure 6: Each strategy removed in turn, and the map re-derived from the remaining submatrix. Left: how far the map moves. Right: how much ε range is lost. The two rank differently, and the difference is the point — the strategy that moves the map most costs no range at all.

Source: `results/fig4_influence.csv`

5 Robustness and threats to validity

Names are not behaviour. An early formulation scored a population cooperative when the strategies named as defectors were extinct. At $\varepsilon \geq 0.16$ on the seven-strategy roster it reported a total victory for cooperation, 0.999, in a population where only 22% of moves were cooperative. The survivor was Grim Trigger, which at any appreciable error rate is tripped within a few rounds and defects thereafter — a defector wearing a cooperative name. All results in this report use the behavioural measure of Section 3.2. That measure verifies itself: across all 354 cells in which Always Defect is the sole survivor, the realised rate equals ε to a mean deviation of 0.0033, which is what it should be when the only cooperative moves left on the board are mistakes.

Right-censoring, and a documented negative result. The grid initially stopped at $\varepsilon = 0.20$, a point at which the pool still cooperates at 0.67. Every reported "ceiling of 0.20" therefore recorded the last column examined and not where any roster broke. This produced a specific false claim: comparing the pool against the pool without Contrite Tit-for-Tat, both read 0.20, and the conclusion drawn was that contrition was sufficient but not necessary — that the mechanism mattered and the strategy did not. Extending the grid to $\varepsilon = 0.35$ gives 0.30 and 0.24 respectively. Contrite Tit-for-Tat is load-bearing. We report the extended estimate and retain the censored one only as a documented negative result. The same censoring made the sensitivity medians unreadable, since a median of 0.20 at every roster size could not mean the median ceiling was 0.20 when 0.20 was the largest value the grid could produce; uncensored they are 0.26 to 0.31. Ceiling statistics now carry an explicit censoring flag in the data and are drawn as bounds rather than points.

The winner was a property of the roster. "Grim Trigger wins" was true of the leaderboard and false as a statement about the strategy (Section 4.1). It survived until the two rows were subtracted opponent by opponent and the entire margin turned out to sit in one column. Every ranking in this report is reported both with and without Random, and the roster sensitivity analysis of Section 4.5 exists because of this result.

Final shares record a transient. The five surviving proportions in the noiseless model look like a result and are a photograph of where the race stopped: the fitness spread among survivors is exactly 0, so selection is not choosing between them. Sampling 1000 starting points confirmed the survivor set as a property of the model and the proportions as almost entirely a property of the start. No proportion is reported here as a finding.

Extinction culling must not feed back into the dynamics. Applying an extinction cutoff inside the update, rather than at reporting time, changes the answer under noise. Forgiving strategies dip below any non-zero threshold during a long non-monotone transient and, once culled, cannot recover. At $\varepsilon = 0.02$, $w = 0.99$ any non-zero cutoff reports "Grim Trigger wins, cooperation 0.213" where no cutoff reports a surviving cooperative mixture at 0.741. All runs here are uncalled during the dynamics.

Bistability, not noise. Twelve of the pool's 152 cells give replicate spreads that no mean describes. At $\varepsilon = 0.10$, $w = 0.7$ the five replicates are 0.10, 0.10, 0.10, 0.45 and 0.83: three runs collapse to the error floor, one reaches a cooperative world and one sits between the two, differing only in the tournament seed. All twelve lie along the boundary, where one expects two attractors to compete — the lowest at $w = 0.5$ and the highest at $w = 0.99$, tracking the edge as it climbs. They are marked in Figure 2 rather than averaged away.

Two replicates in the roster sweeps. The headline maps average five replicates per cell. The leave-one-out influence table and the 120-roster sensitivity sweep re-derive a map for every roster at every cell, so they use two. Both sides of every comparison use the same number, so the comparison is fair and only the error bars widen — but the two counts do not give identical values, and where they differ we have kept each number with its own provenance rather than quietly reconciling them. Removing Contrite Tit-for-Tat costs 0.06 of ceiling at two replicates (Table 1) and 0.08 at five, the decision log's figure. The ranking is unaffected: Soft Majority costs 0.10 under both.

A yardstick for reading map differences. Re-simulating a sub-roster rather than taking its submatrix moves cells by up to 0.054, because the error and continuation draws are keyed by roster position. That figure is the scale at which a difference between two maps stops being readable. It is why Section 4.6 declines to rank the ten- and eight-strategy trims against each other: both move the map by less than a change of random seed would, so the ordering between them is not a measurement.

Convergence. Seventeen of the pool's 760 runs had not settled at $G = 60{,}000$ generations, the latest run to settle doing so at 58,760. They cluster along the boundary — ε between 0.02 and 0.32, w between 0.6 and 0.95 — which is where two competing attractors would slow convergence, and the cell that supplies the bistability example above, $\varepsilon = 0.10$, $w = 0.7$, is among them. They are flagged as provisional in the data rather than presented as settled. (The seven-strategy control roster gives 21 of 440 unsettled with the latest at 59,896; those are the figures the decision log records for it, and they are not the pool's.)

Sampling and payoff level. Twenty trials per matchup was checked against a 500-trial run: no leaderboard position moves and every score shifts by less than 0.01. Under $\varepsilon > 0$ every match is stochastic and trials scale to hold the per-pair round budget constant. On payoffs, negative values are workable at low selection intensity but not guaranteed to be, and the failure is silent — in an ordinary population state under $S = -1$ and $s = 1$ the update assigns negative population shares to three strategies. Non-negative payoffs make the guarantee unconditional, which is why $S = 0$.

Independent verification. All 21 deterministic pairings agree move for move with an established reference implementation [9] over 200 rounds — not merely on score, since comparing totals alone would allow any payoff-neutral behavioural difference through. Stochastic strategies are excluded: two independent generators cannot agree, and comparing distributions would test the generators rather than the engine.

6 Limitations

Our population is well-mixed: every agent is equally likely to meet every other, with no neighbourhoods, no network and no assortment. This is the assumption most likely to matter, since spatial structure is a known route to cooperation that the model cannot represent at all.

No agent learns. Strategies are fixed rules and adaptation happens only through population share. A learned or reinforcement-based strategy might occupy regions of the map that no hand-coded rule reaches.

Only execution error is modelled. Both players always agree about what was played; the misperception case, in which they hold different histories of the same match, is absent. Since Contrite Tit-for-Tat's standing is defined on played moves, it is precisely the strategy whose advantage would be expected to erode under misperception, and that is untested here.

One payoff set carries the main results. We varied $T = 5$, $R = 3$, $P = 1$, $S = 0$ enough to establish that the level does not change the game, but we have not checked the shape of the collapse against a materially different T/R ratio. Everything reported here is one game's geometry.

Three provenance caveats. The sensitivity figure is parsed from a stored run log rather than recomputed, because that sweep takes hours; everything else is regenerated from committed data. The ceiling maxima at roster sizes five to eleven remain censored at $\varepsilon \geq 0.35$ and were not chased further. And of eleven bibliography entries, ten are verified against publisher records or Crossref; Axelrod [1] has no authoritative record — Crossref indexes reviews of the book rather than the book — and its publisher, address and year are marked unchecked in the source file.

7 Conclusion

Cooperation in this model survives far more noise than we would have guessed, and fails far faster. It holds at **0.788** of the achievable band through an error rate of 28% and is effectively gone by 32%. It does not erode in proportion to the error; it holds, and then it snaps. A system observed anywhere below that band looks healthy, and the distance from healthy to gone is two grid steps.

The mechanism is a scissors closing from both sides. As ε rises, retaliators earn less from one another — Tit-for-Tat against itself falls from 3.000 to about 2.26 — while defectors earn more from one another, rising from 1.000 to 1.565, and much more from retaliators, rising from 1.038 to 2.034. The structural advantage that makes reciprocity work is the gap between those quantities, and it loses two thirds of its size across the range. What sustains cooperation past the point where simple reciprocity fails is not more retaliation but less reactivity: mechanisms that decline to treat a single observation as evidence. Of the three the pool contains, dilution is strongest, and its plain-language form is the most transferable sentence in this report — judging someone by their record rather than their last move is the most robust defence against noise the model offers.

How much of that answer belongs to the strategies we chose rather than to the model is measurable, and we measured it. It falls from 0.177 to 0.037 as the roster grows from five to thirteen, and our own seven-strategy roster turns out to have sat at the minimum of 24 draws at its size. A study that stopped at seven would have reported a ceiling less than half the pool's and would have had no way of knowing. We did stop at seven, for a while.

Four times in this project a number that looked like a measurement was a fact about the instrument: a leaderboard reporting its roster, final shares reporting a transient, a name-based metric reporting labels, and a grid reporting its own last column. Each became visible only by returning to what the summary had been computed from — the underlying matrix, the trajectory, the moves played, the range measured.

The fourth differs in kind, and it is the reason this paragraph exists. The first three were errors caught in the model; the fourth was an error in the analysis of the model, made after the same lesson had already been recorded twice in this project's own decision log, and it produced a conclusion that was wrong rather than imprecise. That it recurred is not a lapse to apologise for but the finding: the failure mode is not ignorance of the principle. Every summary is a function of something, and the function has a domain. Where a summary saturates — a ranking with a top, a rate with a floor, a sweep with a last column — values at the boundary stop being measurements and become statements about the boundary, while remaining comparable in appearance.

The two halves of this report are one observation at different scales. A population at $\varepsilon = 0.28$ still cooperating at 0.788 gives no warning that two further steps of noise will take it to 0.162; an indicator holding steady while the thing it measures approaches a cliff is precisely what our four methodological failures were, occurring in the system rather than in the study of it.

We should be clear about the size of what we have shown. This is one game, one population structure, fifteen hand-written rules and no learning; the numbers above are exact for that model and are not measurements of cooperation in the world. What we think travels is not the ceiling of 0.30 but the shape — cooperation holds, and then it snaps. So do the numbers we watch it with.

A Reproduction

Every number in this report derives from files under `results/`, which are committed rather than regenerated on demand. `figures/MANIFEST.md` maps every figure to the file its numbers come from, and each caption carries the same reference.

Randomness derives from a single documented root seed, with per-match generators keyed by tournament coordinates rather than drawn from a running stream, so any individual cell reproduces in isolation and roster changes cannot shift unrelated numbers. Figure PDFs do not embed a creation timestamp, so a rebuild that changes no number produces no diff.

Run `python build_report.py` for the full chain, dominated by the data stage, or `-pdf` to typeset alone. A clean run has been verified end to end and the six derived CSVs returned byte-identical to the committed ones. Full instructions are in the repository README.

B Decision log

`decisions.md` records every methodological choice made during this project, in the order it was made, with the alternatives rejected and why. It is a deliverable rather than a working note: each item in Section 5 is traceable there to the entry that first stated the mistaken version and to the later entry that overturned it. Entries are numbered continuously and are not edited after the fact — where a conclusion was reversed, both the original and the reversal remain.

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